	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** *Academic Year : 2025 – 2026*

Presentation on Capstone Project Execution (CPE–316004)

Topic	Extension Activity Monitoring System for Krishi Vigyan Kendra, Dhule	
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Abstract

- **Krishi Vigyan Kendra (KVK), Dhule** conducts agricultural training programs and extension activities for farmers.
- At present, activity records are maintained manually using registers and spreadsheets, which leads to errors and unorganized data management.
- To overcome these issues, intranet web-based application **"Extension Activity Monitoring System for KVK Dhule"** has been developed.
- It provides a platform to manage Extension activity and training records
- Key features include structured **data entry, Excel import/export, search and filtering, and dashboard-based monitoring**.
- The application supports three user roles: **Programme Coordinator, Subject Matter Specialists, and Programme Assistant**.
- It improves data accuracy, reduces duplication, and enhances transparency and reporting efficiency.
- Developed using the **MERN stack (React, Node.js, Express.js, MongoDB)**.

Content

1. Introduction (background of the industry or used based problem / task)
2. Literature Survey
3. Scope of the project
4. Methodology
5. Details of designs, working and processes
6. Results and applications
7. Conclusion and future scope
8. References and Bibliography

Introduction

- **Krishi Vigyan Kendra** is an agricultural extension center under the **Indian Council of Agricultural Research (ICAR).**
- **KVK Dhule**, established in **1983,** works **under Mahatma Phule Krishi Vidyapeeth, Rahuri** and serves farmers in Dhule district.
- ***Extension*** means spreading agricultural knowledge and new farming techniques from research institutions to farmers through trainings, demonstrations, and workshops.

Discipline	Description
Agronomy	Focuses on crop production, soil management, and field practices.
Horticulture	Deals with cultivation of fruits, vegetables, and flowers.
Home Science	Focuses on nutrition, health, and skill development of rural women.
Plant Protection	Involves pest and disease management in crops.
Animal Science	Deals with livestock management and animal husbandry practices.
Soil Science	Focuses on soil health, fertility, and proper use of fertilizers.

1. Introduction

Krishi Vigyan Kendra (KVK) currently conducts **12 Extension Activities** and **5 Training Programs** to improve farmers' skills, productivity, and farming practices.

Extension Activities (12): Short programs mainly meant to create awareness and share information with farmers.

- Field visits
- Diagnostic Visit
- Group discussion
- Live Web-Casting
- Kisan Mela etc.

Training Programs (5): Structured learning sessions where farmers are taught practical skills and techniques in detail

- Rural Youth Training
- Vocational Training
- Farmers Trainings
- Skill Development Training
- Extension Functionaries training

2.1 Literature Review

1. Study of existing system:

- Studied Krishi Vigyan Kendras (KVKs) – agricultural extension centers under ICAR providing training, technical guidance, and farmer support.
- **731** KVKs across India; **47** in Maharashtra under State Agricultural Universities (e.g., MPKV Rahuri), including KVK Dhule.
- Analyzed how KVK manages extension activities and training programs.
- Key roles involved:
  1. **Programme Coordinator**
  2. **Subject Matter Specialists (SMS)**
  3. **Programme Assistants**
- Programme Coordinator supervises activities, SMS conduct discipline-based work, and Programme Assistant manages data and reports.

2.1 Literature Review

The following points were discussed with each of the employee who are responsible for managing this activites.

1. Programme Coordinator

- How many stakeholders are involved in KVK activities?
- What are your main responsibilities at KVK?
- What type of decisions are made based on this data?
- Why are these records maintained?
- What challenges do you face in monitoring activities?
- How are activity reports submitted to you?
- How are extension activities and training programs planned?
- Do you face delays in getting information?

Literature Review

2. Subject Matter Specialist(SMS):

- Does every discipline have a separate SMS?
- Do all SMS perform similar roles?
- Is any other staff involved with you while planning or conducting activities and training?
- What type of activities and training programs are conducted at KVK Dhule?
- How do you know which activities you have to conduct?
- Are the activities pre-planned?
- Do you follow any schedule for activities?
- How frequently are activities conducted?
- How do you record extension activities?
- What information do you maintain for each activity?

Literature Review

3. Programme Assistant:

- Do you record the data for all disciplines? How do you collect activity information?
- How is the data stored digitally?
- How is the data stored digitally?
- What information do you record into Excel?
- How is funding for activities managed or recorded?
- How do you organize activity-wise data?
- What challenges do you face during data entry?
- If any register is lost, does it create any problem? What kind of problem?
- Do you face issues while managing large data?

Literature Review

1. Problem identification:

There are following points that indicates the need of system to overcome this difficulties

i. To develop a web-based intranet system for monitoring extension activities and training programs

ii. To enable centralized data collection and storage

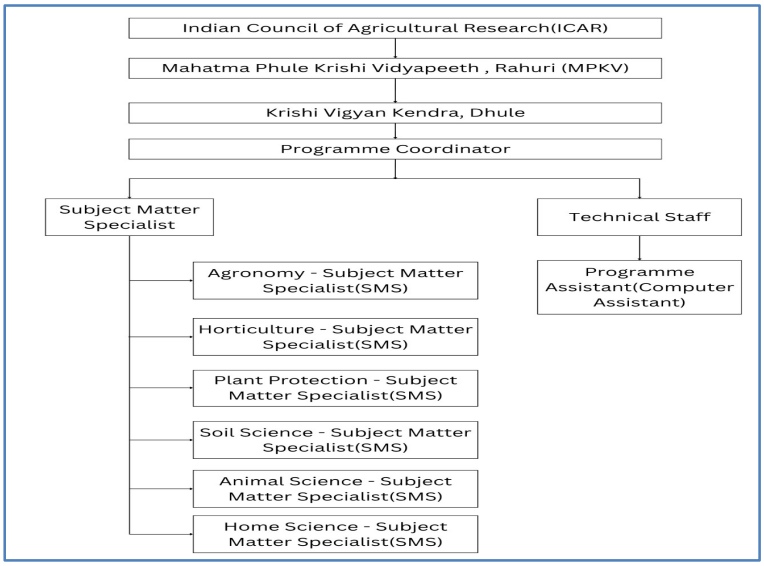
iii. To reduce manual effort and eliminate data duplication

iv. To improve data management and easy accessibility of information

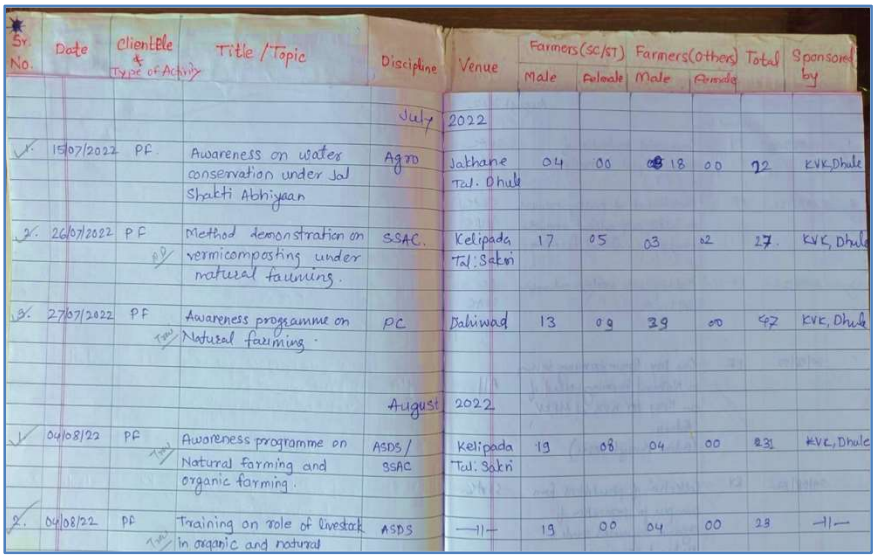
v. To support data analysis for better decision-making

vi. To generate reports in both tabular and graphical formats

Literature Review



Literature Review



Sr. No.	Date	Clientele & Type of Activity	Title / Topic	Discipline	Venue	Farmers (SC/ST) Male	Farmers (SC/ST) Female	Farmers (Others) Male	Farmers (Others) Female	Total	Sponsored by
				July	2022						
1.	15/07/2022	PF	Awareness on water conservation under Jal Shakti Abhiyaan	Agro	Jakhane Tal. Dhule	04	00	18	00	22	KVK, Dhule
2.	26/07/2022	PF	Method demonstration on vermicomposting under natural farming.	SSAC.	Kelipada Tal. Sakri	17	05	03	02	27.	KVK, Dhule
3.	27/07/2022	PF	Awareness programme on Natural farming.	PC	Dahiwad	13	02	39	00	57	KVK, Dhule
				August	2022						
1.	04/08/22	PF	Awareness programme on Natural farming and organic farming.	ASDS / SSAC	Kelipada Tal. Sakri	19	08	04	00	31	KVK, Dhule
2.	04/08/22	PF	Training on role of livestock in organic and natural	ASDS	—"—	19	00	04	00	23	—"—

Literature Review

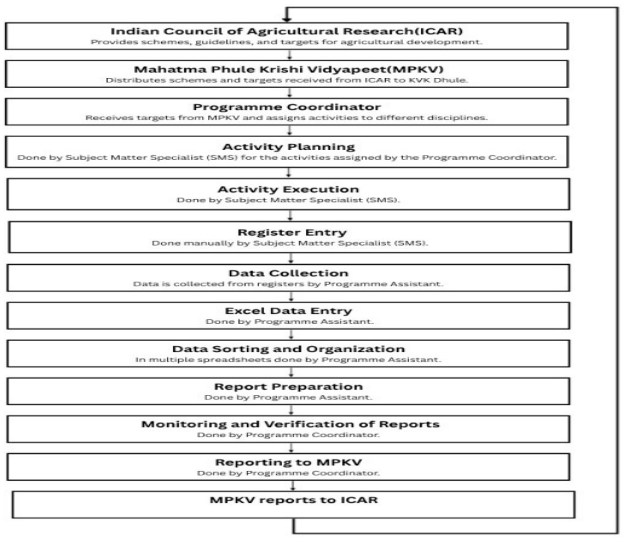
Event Name/Sub Category	Start Date	End Date	Venue	Target Group	Contact Person		SC/ST		Other		EF	
							Male	Female	Male	Female	Male	Female
Field day on feeding of moringa leaves on growth performance of goat kids	8/5/2024	8/5/2024	Chaugaon Tal: Dhule	Farmer	Dr. D. M. Choudhary		56	0	4	0	1	0
Field day of Shri. Dada Lad technology Bt. Cotton plot under Special Cotton Project	25-09-2024	25-09-2024	Betawad Tal: Shindkheda	Farmers and Farm Women	Mr. Jagdish Kathepuri.		8	0	81	3	5	0
Field day of FLD Soybean under Oilseed Model Village	8/10/2024	8/10/2024	Bopkhel Tal: Sakri	Farmers and Farm Women	Dr. D. M. Choudhary, Shri. J. V. Kathepuri		112	2	23	0	6	0
Field day of CFLD Pigeon pea plot	11/10/2024	11/10/2024	Nyahalod Tal: Dhule	Farmers and Farm Women	Dr. Pankaj Patil, Shri. Jagdish Kathepuri		8	2	92	0	5	0
Field day of CFLD Soybean under CFLD Oilseed	16-10-2024	16-10-2024	Kurkhali Tal: Shirpur	Farmer	Dr. Pankaj Patil, Shri. Jagdish Kathepuri		4	0	68	0	3	0
Field day of Shri. Dada Lad Technology Bt. Cotton plot under Special Cotton Project	6/12/2024	6/12/2024	Dighave Tal: Sakri	Farmer	Mr. Jagdish Kathepuri		19	0	80	0	3	0
Field day of Shri. Dada Lad technology of Bt Cotton plot under Special Project on Cotton	13-12-2024	13-12-2024	Bhadane Tal: Shindkheda	Farmer	Mr. Jagdish Kathepuri		13	0	84	0	4	0
Field day of Bt Cotton plot under Special Project on Cotton	19-12-2024	19-12-2024	Borvihir Tal: Dhule	Farmer	Mr. Jagdish Kathepuri		6	0	46	0	4	0
						8	226	4	478	3	31	0

Sheet1 | Sheet3 | Field visit | Diagnostic visit | Other extension | Sheet2 | Method demo | Sheet4 | Kisan Goshti | Group discussio

Each field data need to be entered manually which is time consuming and prone to errors.

Existing records stored in multiple sheets. Data divided into multiple sheets as per the activity name and common table maintained for all the discipline data

Literature Review



2.2 Problem Definition

- Problem Definition

At Krishi Vigyan Kendra (KVK), Dhule, extension activities and training data are managed using manual registers and multiple Excel sheets, making the process time-consuming and unorganized. This leads to difficulties in tracking activities, data duplication, and errors. Therefore, there is a need to develop a web-based intranet Extension Activity Monitoring System that enables centralized data storage, efficient management, better analysis for decision-making, and easy report generation in tabular and graphical formats.

3.1 Scope of the project

➢ **Functional Scope**

- The system allows adding, updating, and deleting activity records.
- It stores details like activity name, date, venue, discipline, and participants.
- It manages different types of activities such as training, workshops, and field visits.
- It provides role-based access for Admin, SMS, and Programme Assistant.
- It enables easy search and filtering of records.
- It generates reports in tabular and graphical formats.
- It maintains centralized data for all disciplines.

Scope of the project

➢ **Non-Functional Scope**

- The system provides a user-friendly and simple interface.
- It ensures fast data access and quick response time.
- It maintains secure login with controlled user access.
- It ensures reliable data storage with minimal errors.
- It is easy to maintain and update.
- It works within an intranet environment without internet.
- It is scalable for future data and user growth.

3.2 System Requirement

➢ **System Requirements**

- Operating System: Windows
- Platform: Visual Studio Code
- Languages: React.js, HTML, CSS, JavaScript
- Backend: Node.js, Express.js
- Database: MongoDB

➢ **Hardware Requirements**

- Processor: Intel i3 Processor Core
- Hard Disk: 128 GB (minimum)

3.3 Users of the System

1. Programme Coordinator / Head

- **Roles**
  - Approve or reject SMS registration requests
  - Add and manage disciplines
  - Add activities and training programs
  - Assign disciplines to SMS
  - Provide permissions such as add, edit, update, delete, and import data
  - View analytics, graphs, and overall data of all disciplines
  - Perform operations like searching and sorting records

Users of the System

2. Subject Matter Specialist (SMS)

- **Roles**
  - Register in the system
  - Manage extension activities and training data for the assigned discipline
  - Perform operations such as add, edit, update, delete, and import data based on permission
  - View analytics for their assigned discipline
  - View data of other disciplines in read-only mode

Users of the System

3. Programme Assistant

- **Role**
  - Enter and manage activity and training data for all disciplines
  - Maintain and update digital records in the system
  - View analytics and reports for all disciplines
  - Assist in data verification and record management

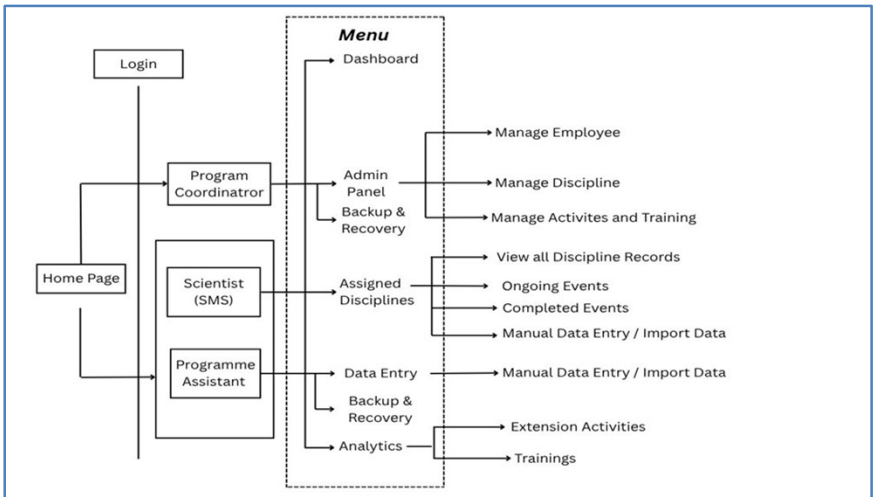
4. Methodology

- **Requirement Gathering and Analysis**
  - Requirements were gathered and analyzed to understand user needs, data handling, and required functionalities for the Extension Activity Monitoring System.
  - The system includes three users: Programme Coordinator, Subject Matter Specialists (SMS), and Programme Assistant, each responsible for planning, conducting, monitoring, and managing activity data.
- **Study of Existing System**
  - Studied the current process of maintaining records in registers and Excel sheets.
  - Identified difficulties in managing, searching, and tracking activity records.
- **Development Model (Waterfall Model)**
  - The Waterfall Model was followed to develop the system step by step.
  - The system was built in modules and improvements were made during development.

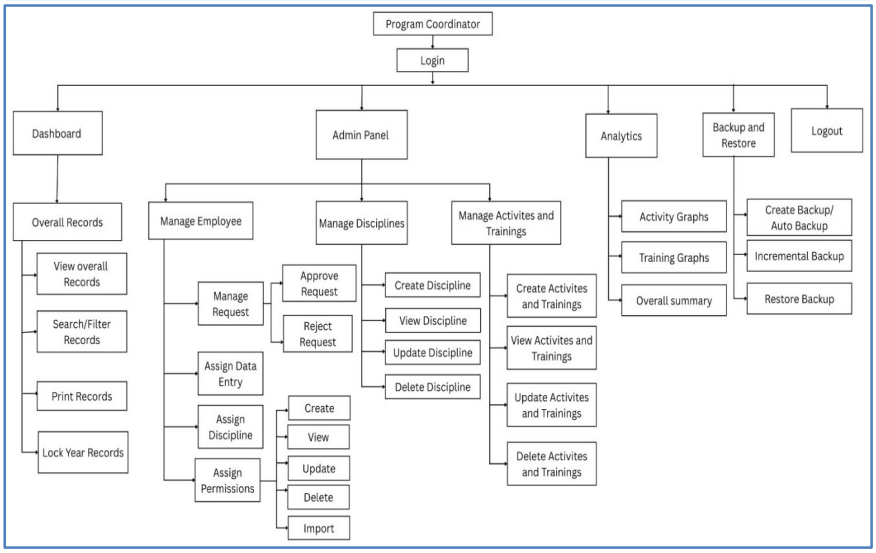
Methodology

- **System Design**
  - Planned the overall structure of the application.
  - Divided the system into role-based modules (Admin, SMS, Computer Assistant).
  - Designed the workflow of registration, approval, discipline assignment, and data entry.
- **System Development**
  - Developed the application using the MERN Stack (MongoDB, Express.js, React.js, Node.js).
  - Implemented modules for user registration, admin approval, discipline management, and activity data management.

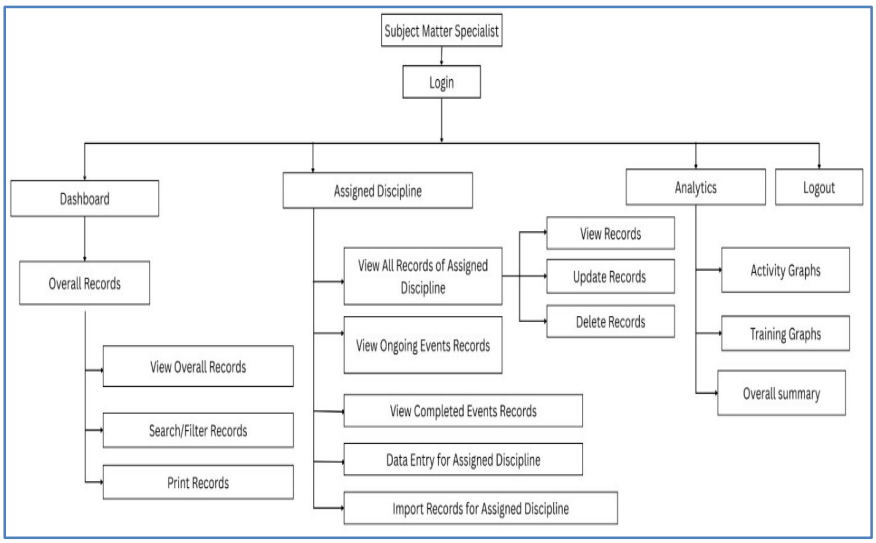
Menu Diagram



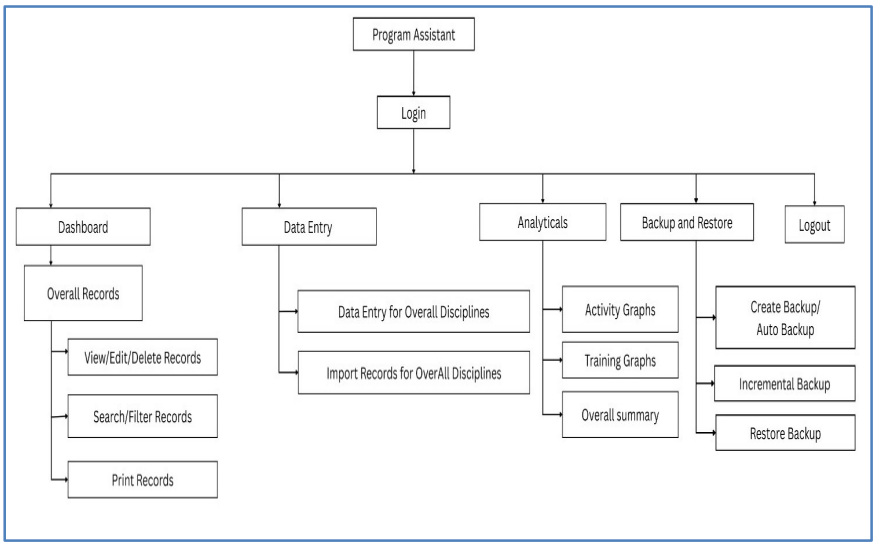
Programme Coordinator



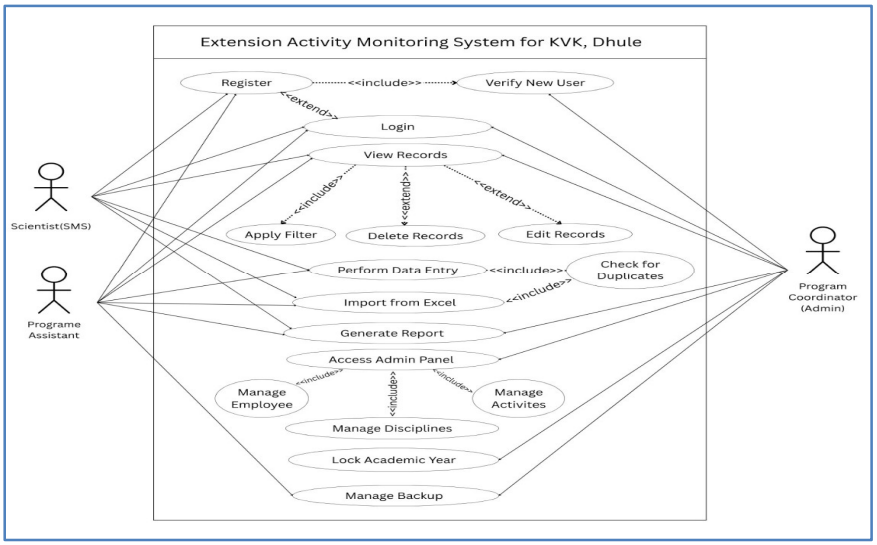
Subject Matter Specialist



Programme Assistant



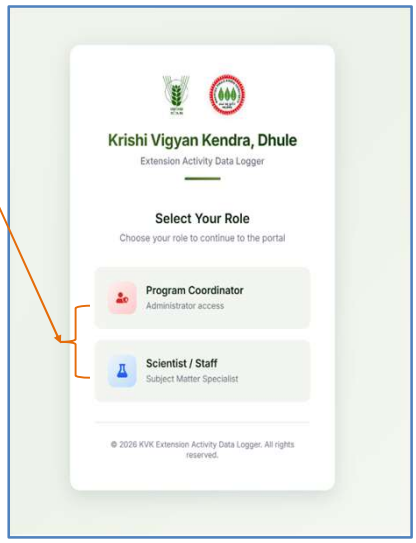
Use Case Diagram



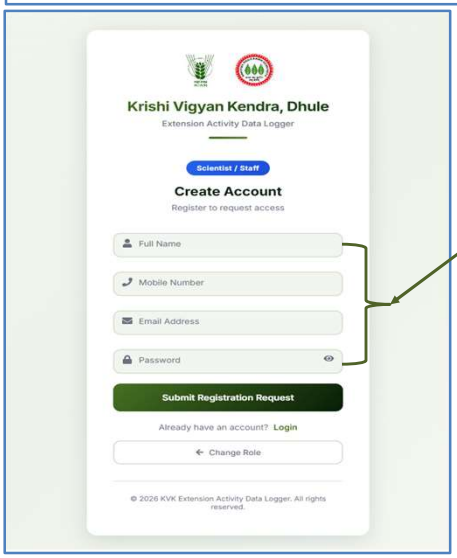
5. Details of designs, working & processes

Role-based authentication means users can access the system based on their assigned role. It helps control what actions each user can perform and ensures secure access to the system.

➢ **Users of the System:**
- Programme Coordinator
- Subject Matter Specialist(SMS)
- Programme Assistant



5. Details of designs, working & processes



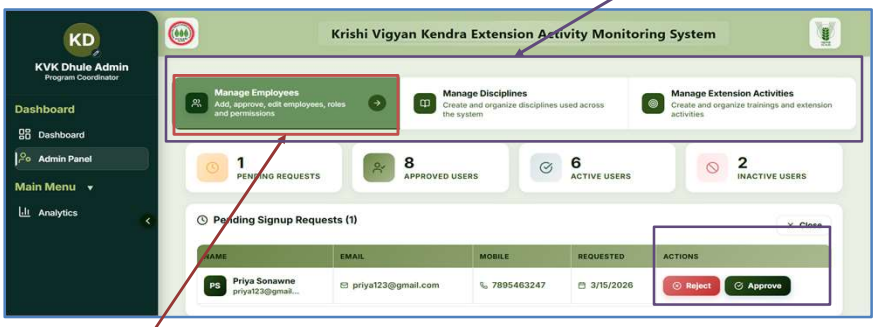
Scientist registration form where the SMS enters details such as **Name, Email, Mobile Number, and Password** and submits the request. The request is then sent to the admin for approval.

If the admin approves the request the user can login else the user has to register again.

5. Details of designs, working & processes

1. Flow and Design For Admin Panel

Admin navigation bar with options like **Manage Employees, Manage Discipline, and Manage Extension Activities** for managing users, disciplines, and activities.

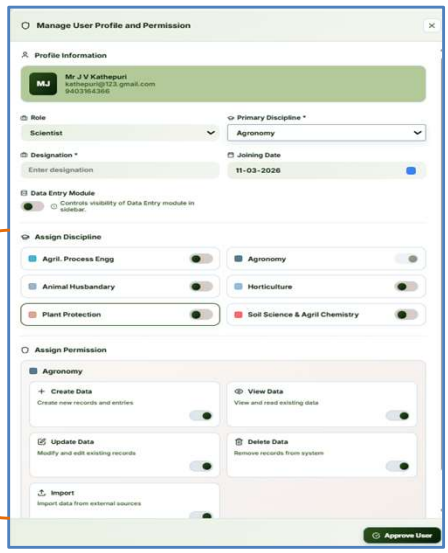


Admin panel: **Manage Employee** where registration requests are received. The admin can approve or reject the request.

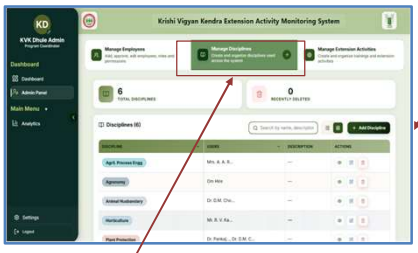
5. Details of designs, working & processes

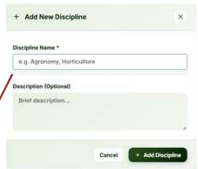
Request Rejected: Admin enters the reason and password, and the user is rejected.

Request Approved: Admin assigns discipline and permissions to the user.

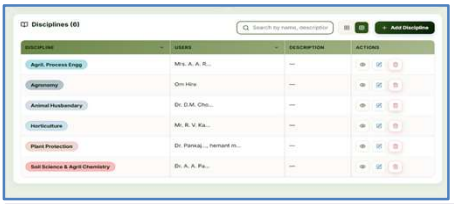


5. Details of designs, working & processes





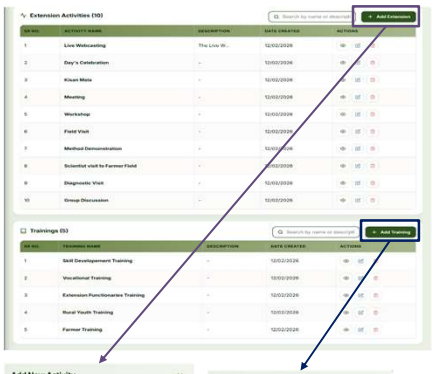
Admin can create new discipline by clicking on add discipline button

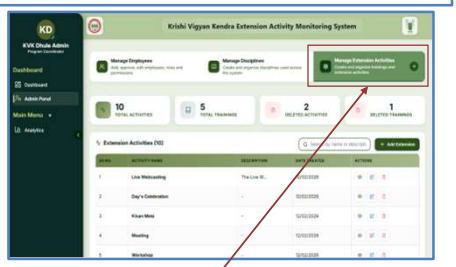


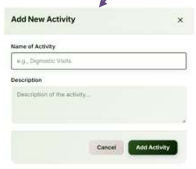
Manage Discipline Module
From this module, the admin can create new disciplines. These disciplines can be assigned to employees (Scientists), and the assigned discipline will appear in the employee dashboard sidebar.

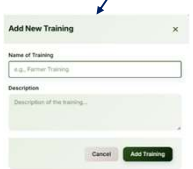
The discipline added are shown in the table with discipline name user assigned to it. Admin can also edit view and delete discipline

5. Details of designs, working & processes









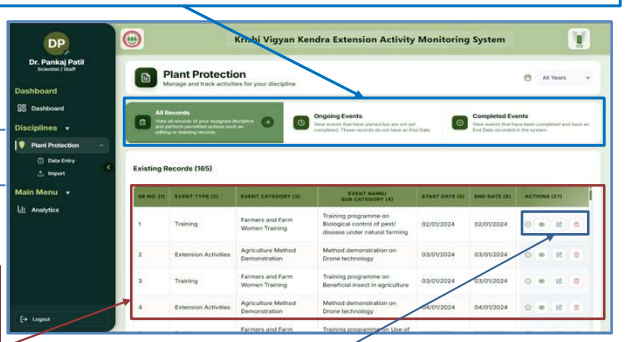
Manage Activities Module
From this module, the admin can **add and manage extension activities and trainings**. These activities are stored in the system and are used while entering activity records.

Admin can add activities and trainings by clicking on button add extension and add add trainings the added trainings and activities will be displayed in their respective tables

5. Details of designs, working & processes

It is divided into three panels: **All Records**(all records including ongoing as well as completed), **Ongoing Events**(Event records which have been generated but not yet completed), and **Completed Events**.(Records which were scheduled and are completed-records with start and end date)

Scientist(SMS) will have access only to the discipline assigned to them they can do manual data entry and excel data import (only if permission is given) for there particular discipline only



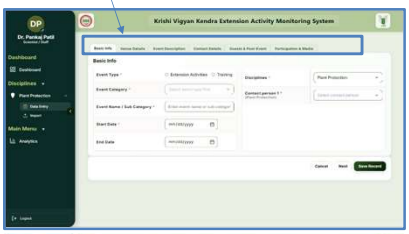
The table will contain overall of the particular discipline which have been assigned to the Scientist (Ongoing Records and Completed Records)

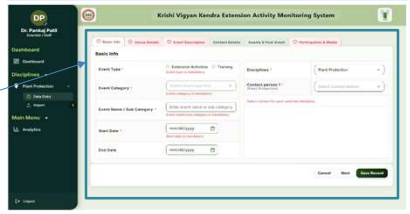
User has the permission that has been assigned by the Admin to perform on data such as edit delete

5. Details of designs, working & processes

The entire form is divided into different sections to make data entry easy and organized. Sections such as Basic information, venue details, Event Description, Contact Details, Guest and post event, participation and media

Data Entry Form
Used to enter event details, divided into sections for easy use.
Basic Info – Type, category, name, dates, discipline, contact.
Venue – Place, taluka, district.
Description – Objectives, details, target group.
Contact – Organizer details.
Guest – Chief guest and others.
Participation & Media – Participants and media.





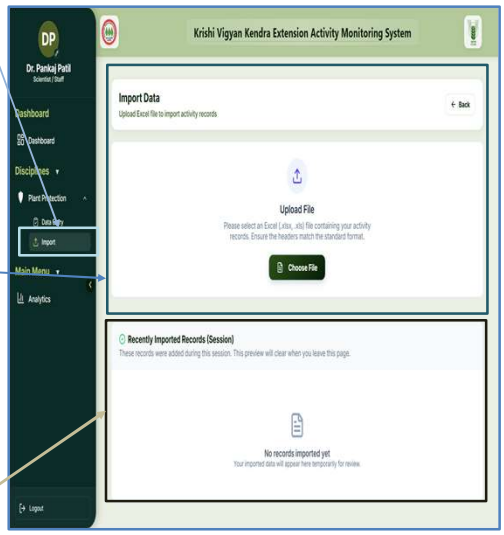
All required fields must be filled to ensure accurate data entry. After saving, the record is stored in the system and displayed as either **All Records** (with start and end date) or **Ongoing Records** (with only start date) under the admin's discipline.

5. Details of designs, working & processes

Import Module : For importing excel records

If the user has data entry permission, they can use the **Import Module** to upload Excel files of previous years, allowing them to store records in the system from 2016 onwards.

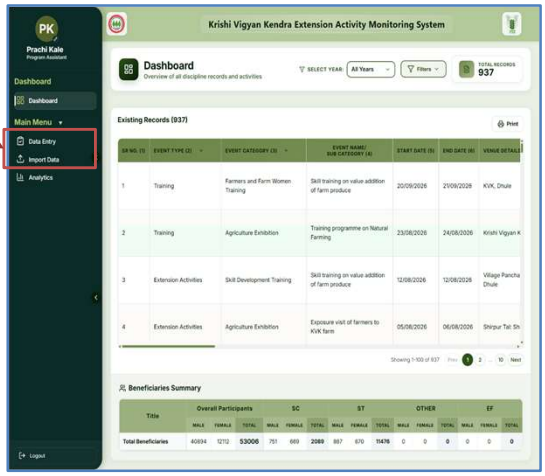
Records imported through the module are displayed in this section for a limited time. They remain visible until the user refreshes or navigates to another page, after which they are cleared from the view.



5. Details of designs, working & processes

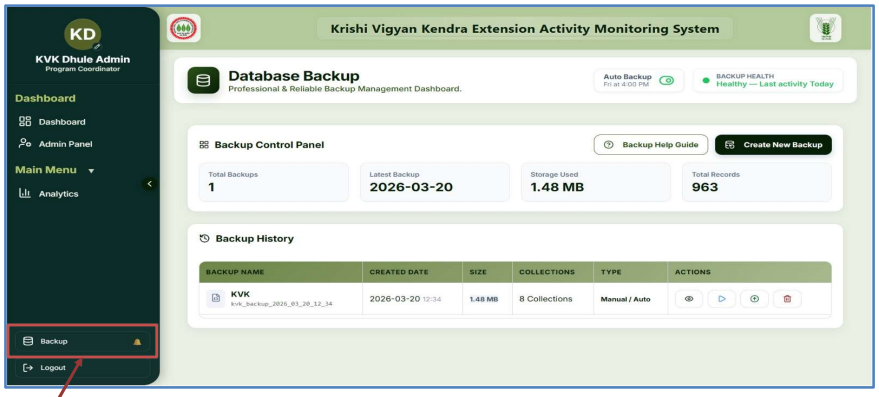
3. Flow and Design For Computer Assistant Panel

The **Computer Assistant** is not assigned to any specific discipline and therefore has permission to perform data entry and data import across all disciplines in the system.



5. Details of designs, working & processes

3. Flow and Design For Computer Assistant and programe coordinatorPanel



- The system provides **manual and automatic data backup** for the **Programme Coordinator and Programme Assistant**.
- Automatic backup runs **every Friday (9 AM – 5 PM)** to ensure data safety.

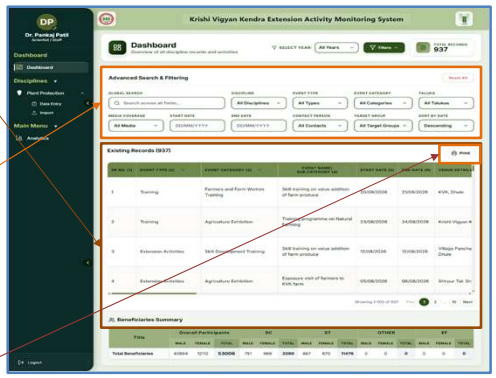
5. Details of designs, working & processes

- Common features for Admin, Scientist and Computer Assistant

The Dashboard Module will have the overall records of the system. This feature will be same for every user in the system.

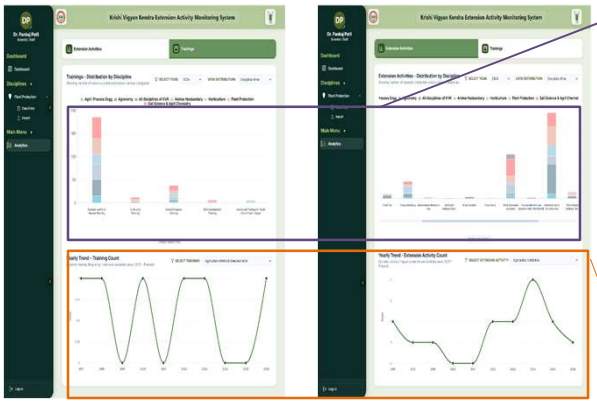
A search filter panel has been provided to allow users to search records as per desired. Results can be searched and sorted based on various fields of the table

It will include print function for printing the records



5. Details of designs, working & processes

The **Analytical Module** is provided to:
Track performance – Monitor number and type of events conducted

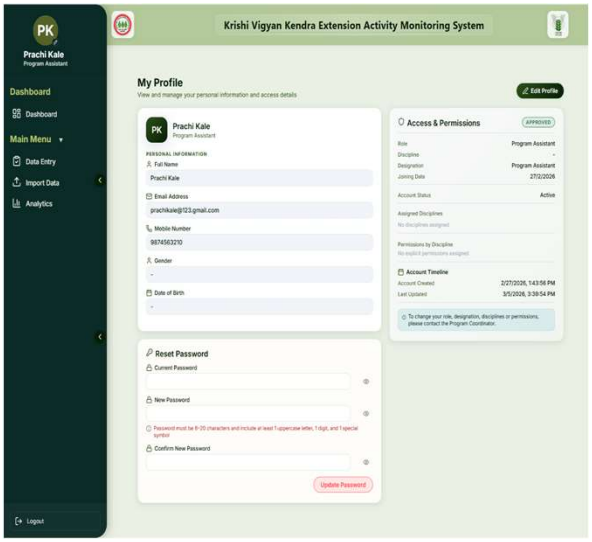


Extension Activities – Distribution by Discipline
Shows the count of total number of particular events or trainings conducted in each discipline

Yearly Trend – Extension Activity Count
Shows the number of activities conducted each year from 2017 to the present, helping to understand the increase or decrease in activities over time.

5. Details of designs, working & processes

The **User Profile** page shows users personal details and account information. You can view and update your info, check your role and permissions, and change your password here.



6. Applications

➢ **Applications of the System:**

- Manages extension activities and training programs efficiently at Krishi Vigyan Kendra.
- Maintains organized digital records of all activities and trainings in one place.
- Enables easy data entry and management by scientists and staff.
- Allows quick search, filtering, and sorting of records.
- Generates reports and basic analytics to support better decisions.
- Improves tracking and monitoring of activities across different disciplines.

7. Conclusion

The Extension Activity Monitoring System for KVK Dhule is a web-based application designed to manage and monitor extension activities and training programs in a simple and organized manner. It provides a centralized platform to store and manage all activity records, making data entry, updating, and access easier for staff and scientists. The system reduces manual effort, improves data accuracy, and ensures better organization of records. It also supports quick searching, filtering, report generation, and graphical analysis, helping in effective monitoring and informed decision-making.

7. Future Scope

- Introduce flexible and well-structured report formats for better data presentation.
- Add bulk operations (update, delete) to manage large data efficiently.
- Implement advanced filtering and customized report generation options.
- Strengthen data validation to minimize errors and improve accuracy.
- Provide additional fields and customization for future adaptability.Enhance user interface for a more simple and user-friendly experience.Improve system performance and speed for handling large datasets.Enable better accessibility and scalability for long-term use.

8. References and Bibliography

- **KVK Dhule Website** (https://kvkdhule.icar.gov.in): Provides details about extension activities, training programs, and services for farmers.
- **ICAR Website**(https://icar.org.in): Offers information on agricultural policies, schemes, and the role of ICAR in guiding KVKs.
- **MPKV Website** (https://mpkv.ac.in): Explains agricultural education, research, and supervision of KVKs by MPKV.
- **Software Engineering & Testing Books**: Used for understanding system development and testing concepts.
- **KVK Dhule Staff Interaction**:
  - **Programme Coordinator** explained planning, monitoring, and reporting of activities.
  - **Subject Matter Specialist** shared insights on conducting activities and maintaining records.
  - **Programme Assistant** explained data entry, report generation, and challenges in manual systems.

Any Questions?